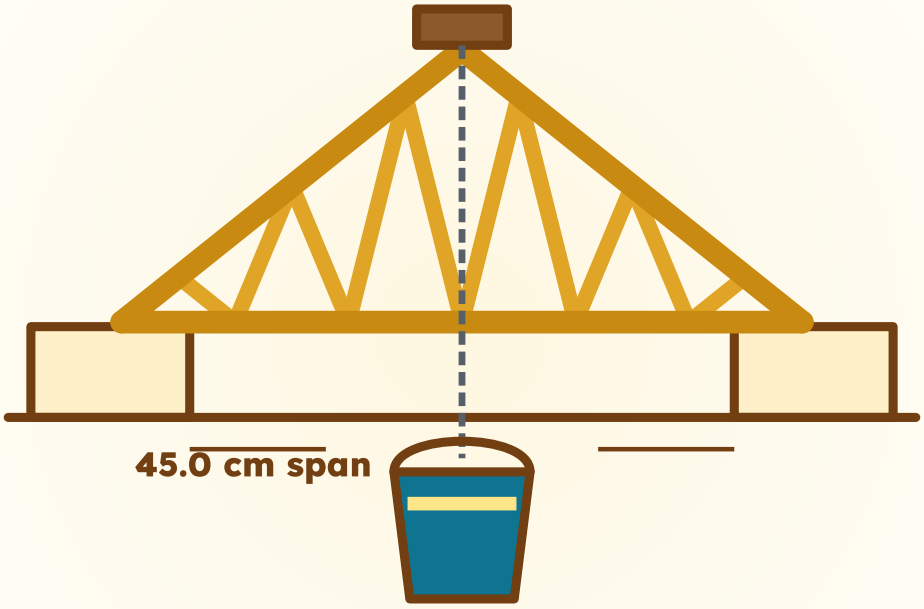
ENGINEER'S Logbook

Plan it. Build it. Test it. Make it better.



ENGINEER ID

Name: _____

Partner: _____

Event: Bridge-A-Roni

If found, please return to the DDMES Science Olympiad binder shelf.

How an engineer works

Real engineers keep a **logbook**. They write down every plan, every build, every test, and every break. This is yours. Bring it to every build session and keep it with your bridge at home.

1
PLAN
Draw your bridge on graph paper with real numbers.

2
BUILD
Pasta and glue only. Small dabs. Let it dry.

3
TEST
Mass it, load it, watch where it breaks.

4
LOG
Write every number right away in a Trial Log.

5
IMPROVE
Change ONE thing. Build Version 2. Beat your score.

The Engineer's Code

- [ ] Safety glasses on whenever pasta is snapping or sand is pouring.
- [ ] **You** build. Grown-ups watch, hold a ruler, and ask questions.
- [ ] Only pasta and glue. No tape, paint, string, or rubber bands on the bridge.
- [ ] Small glue dabs: extra glue adds mass and lowers your score.
- [ ] Change one thing at a time so you know what made the difference.
- [ ] The break is not a failure. The break is data!

My build kit

- [ ] This logbook and a pencil
- [ ] Spaghetti (and a little linguine for the platform)
- [ ] White glue or wood glue, paper plate, toothpicks
- [ ] Wax paper, painter's tape, push pins
- [ ] Building board and your full-size plan
- [ ] Ruler; the team's 0.01 g scale
- [ ] Safety glasses

Journal map

Know your event	p. 3-4
Plan and design	p. 5-7
Build log and Trial Logs	p. 8-10
Graph and fix it	p. 11-12
At-home challenges	p. 13-15
Tournament, missions, badges	p. 16-18

Where engineers find ideas

Real bridges
Look for triangles on bridges, power towers, and cranes around town.

Your broken bridge
Where it snapped tells you what to make stronger.

The NCSO page
Videos of pasta bridges being built and tested.

The playground
Swing sets and climbers use triangles and braces.

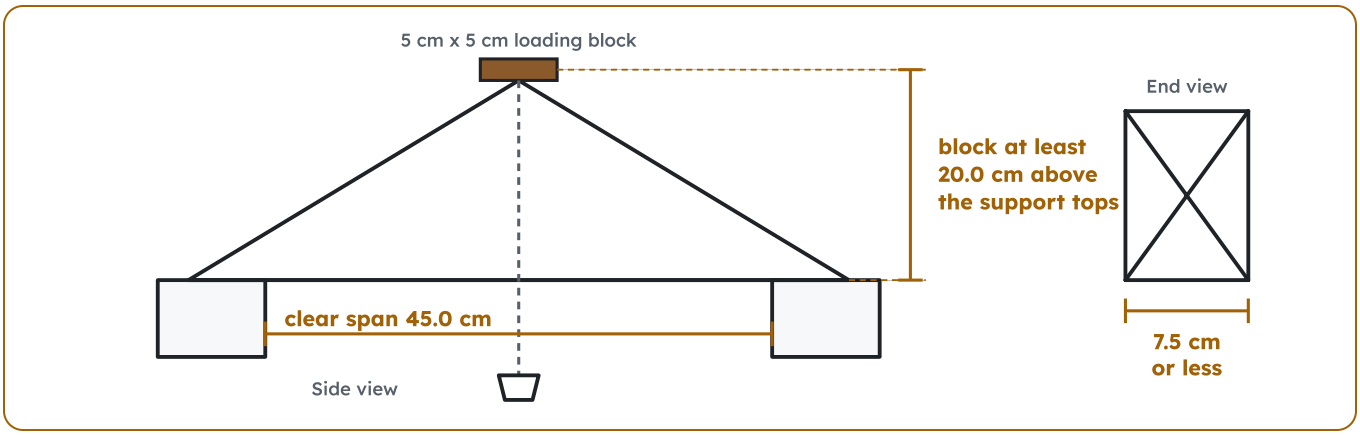
Your partner's truss
Lay the two trusses side by side: they must match.

The score sheet
Check it before every test so you stay in Tier 1.

Did you know? A bundle of 4 spaghetti strands can hold much more push than 4 single strands used one at a time. Bundles are how pasta bridges hold thousands of grams!

Know Your Event: Rules and Scoring

You will build a bridge out of only pasta and glue that is as light as possible but holds as much weight as possible. The winner is the bridge with the best structural efficiency: the load it holds divided by its own mass.



The rules in numbers

Materials	Only pasta and glue. No tape, paint, string, or rubber bands.
Clear span	45.0 cm between the supports. Make the bridge longer so its ends rest on top of them.
Block height	The 5 cm x 5 cm loading block must sit at least 20.0 cm above the tops of the supports.
Width	No wider than 7.5 cm.
Opening	Leave a gap under the block for the bolt and chain.
Max load	10 kg (10,000 g). Safety glasses on the whole time.

How you win

Efficiency = load held (g) ÷ bridge mass (g)

- Bigger is better. **Example:** a 50 g bridge holds 5,000 g. 5,000 ÷ 50 = **100**.
- A load over 10 kg counts as **10,000 g**.
- **Tier 1** passes every score sheet check. Every Tier 1 bridge beats every Tier 2 bridge.
- Same efficiency? The **lighter** bridge wins.

Test day: 10 minutes for you and your partner

1. Glasses ON. Bridge is massed to 0.01 g and measured.
2. Set the bridge so its ends rest only on the support tops.
3. Set the loading block on the center. The chain hangs through the gap.
4. Hang the bucket. It must not touch the floor.
5. Pour sand for up to 3 minutes. Steadier keeps the bucket still.
6. Stop at a break, at 10 kg, or at 3 minutes.

Load held = loading block + chain + bucket + sand, weighed on a big scale. Sand poured after the break does not count.

Practice: score it

1. Bridge A is 60 g and holds 6,000 g. Bridge B is 30 g and holds 4,500 g. Which wins? Show your math.

2. A 40 g bridge holds 12,000 g. What load is used for the score? What is the score?

Check: 1) A = 100, B = 150, so Bridge B wins. 2) The load counts as 10,000 g (the max). 10,000 ÷ 40 = 250.

Know Your Event: Forces and Trusses

Science words

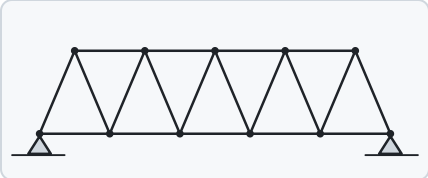
Force	A push or a pull.
Tension	A pull that stretches a piece.
Compression	A push that squeezes a piece.
Truss	A frame made of triangles.
Structural efficiency	Load held (g) divided by bridge mass (g). Bigger is better.

Push or pull?

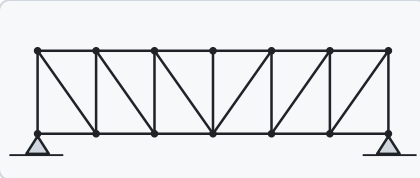
Pasta is **strong when pulled** (tension) but a long strand **bends and snaps when pushed** (compression). Short pieces and bundles hold much more push.

Triangles win: a triangle cannot change shape unless a side breaks. A square folds at its corners. One diagonal turns a square into 2 triangles.

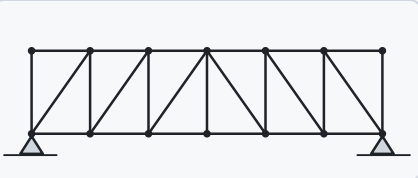
Three truss patterns



Warren
zig-zag diagonals, no posts

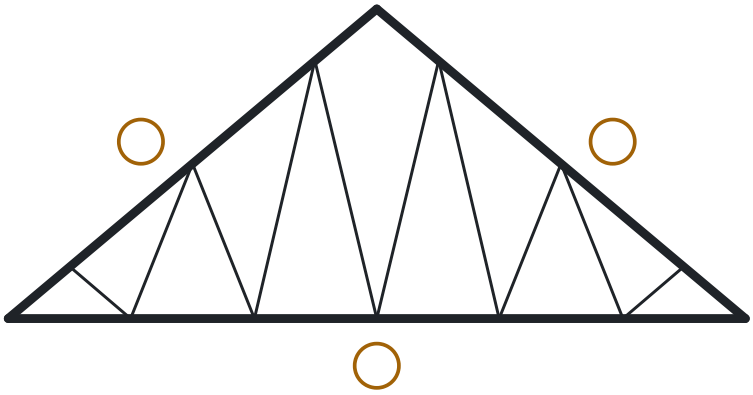


Pratt
posts; diagonals slant DOWN to the middle



Howe
posts; diagonals slant UP to the middle

Force Finder: label our starter truss



bottom chord

Write **T** (tension) or **C** (compression) in each circle.

Hint: the bottom chord keeps the two feet from sliding apart.

My prediction: the piece that will break first is the

_____

_____

☐ Force Finder badge: all 3 circles right

Spaghetti test

What I did (glasses on!)	What happened?
Pulled one strand (tension)	
Pushed the ends of one long strand (compression)	
Pushed a short 5 cm piece	
Pushed a bundle of 4 strands	

Check: the two slanted sides are C (pushed). The bottom chord is T (pulled).

Plan It: My Mass Budget

Before you build, estimate how heavy your bridge will be. Lighter bridges score higher, so every strand counts! (This matches Independent Lab I2.)

Estimate facts

One strand	A 25 cm strand of spaghetti is about 1 g.
Glue	Glue adds a few grams. Small dabs keep it low.
Starter plan	Bottom chord 50 cm. Each slanted side about 33 cm. All diagonals together about 150 cm.
Tier 1	Passes every score sheet check. Any Tier 1 bridge beats every Tier 2 bridge.
Max load	A load over 10,000 g counts as 10,000 g.
Tie	Same efficiency? The lighter bridge wins.

Measure with string

Lay a string along each line of your plan, then measure the string with a ruler. Multiply by the strands in that bundle.

> **Starter plan with 3-strand chords and sides:** one truss uses about 500 cm of pasta. That is about 20 strands, or 20 g.

Mass budget (one side truss)

Part	Length (cm)	Strands in bundle	Pasta used (cm)
Bottom chord			
Slanted side 1			
Slanted side 2			
All diagonals		1	
Total for one truss			

Budget math

1. Two trusses: total x 2 = _____ cm. Add 150 cm for braces and platform = _____ cm.

2. Divide by 25 to get strands. My bridge will be about _____ g (add a few grams for glue).

3. My efficiency goal is _____. So my bridge must hold: mass _____ x goal _____ = _____ g.

Estimate vs. real

	My estimate (g)	Real mass (g)	How close?
Version 1			
Version 2			

☐ Mass Budgeter badge: budget done and compared to the real mass.

Design Sketch: Version 1

Draw your first bridge before you snap any pasta. Use the starter plan: 50 cm bottom chord, top point 21 cm above the middle, diagonals about every 8 cm.

Date _____ **Truss pattern** _____ **Strands per bundle** _____

SIDE VIEW: draw one side truss. Mark the 45.0 cm span and the platform.

Scale: 1 small square = 2 cm (so the 50 cm bottom chord is 25 squares)

END VIEW

Measurement	Rule or tip	My plan
Bottom chord length	50 cm	
Platform height	20.0 cm or more (aim 20.5-21)	
Outside width	7.5 cm or less	
Gap for bolt and chain	about 3 cm	
Estimated mass	from p. 5	

The ONE big idea in this version:

Design Sketch: Version 2

After your test, redraw your plan. Show your ONE change in a different color. Then build it at full size on wax paper.

Date _____ **Truss pattern** _____ **Strands per bundle** _____

SIDE VIEW: draw one side truss. Mark the 45.0 cm span and the platform.

Scale: 1 small square = 2 cm (so the 50 cm bottom chord is 25 squares)

END VIEW

Measurement	Rule or tip	My plan
Bottom chord length	50 cm	
Platform height	20.0 cm or more (aim 20.5-21)	
Outside width	7.5 cm or less	
Gap for bolt and chain	about 3 cm	
Estimated mass	from p. 5	

The ONE big idea in this version:

Build Log

Every time you build (at a session or at home), write one line. Work 15 to 20 minutes at a time and let glue dry a day between steps. Mass the bridge with the team's 0.01 g scale when you can.

Date	Minutes	What I built or fixed	Mass (g)	Grown-up initials

☐ Truss Builder badge: side truss finished at home, every joint dry and tight.

Trial Log

Trial #1 Date _____ Version _____ ☐ practice ☐ official

What I changed (just ONE thing):

_____

_____

My prediction: it will hold _____ g and break at the…

_____

_____

Bridge mass (to 0.01 g)	
Width (cm) / platform height (cm)	
Tier (1 or 2)	
Why it stopped: broke, 10 kg, or 3 min	
Load held (g), 10,000 g max	
Efficiency = load ÷ mass	

Where did it break? Sketch it.

What I learned:

_____

_____

_____

Next time I will change:

_____

_____

Trial #2 Date _____ Version _____ ☐ practice ☐ official

What I changed (just ONE thing):

_____

_____

My prediction: it will hold _____ g and break at the…

_____

_____

Bridge mass (to 0.01 g)	
Width (cm) / platform height (cm)	
Tier (1 or 2)	
Why it stopped: broke, 10 kg, or 3 min	
Load held (g), 10,000 g max	
Efficiency = load ÷ mass	

Where did it break? Sketch it.

What I learned:

_____

_____

_____

Next time I will change:

_____

_____

Trial Log

Trial #3 Date _____ Version _____ ☐ practice ☐ official

What I changed (just ONE thing):

_____

_____

My prediction: it will hold _____ g and break at the…

_____

_____

Bridge mass (to 0.01 g)	
Width (cm) / platform height (cm)	
Tier (1 or 2)	
Why it stopped: broke, 10 kg, or 3 min	
Load held (g), 10,000 g max	
Efficiency = load ÷ mass	

Where did it break? Sketch it.

What I learned:

_____

_____

_____

Next time I will change:

_____

_____

Trial #4 Date _____ Version _____ ☐ practice ☐ official

What I changed (just ONE thing):

_____

_____

My prediction: it will hold _____ g and break at the…

_____

_____

Bridge mass (to 0.01 g)	
Width (cm) / platform height (cm)	
Tier (1 or 2)	
Why it stopped: broke, 10 kg, or 3 min	
Load held (g), 10,000 g max	
Efficiency = load ÷ mass	

Where did it break? Sketch it.

What I learned:

_____

_____

_____

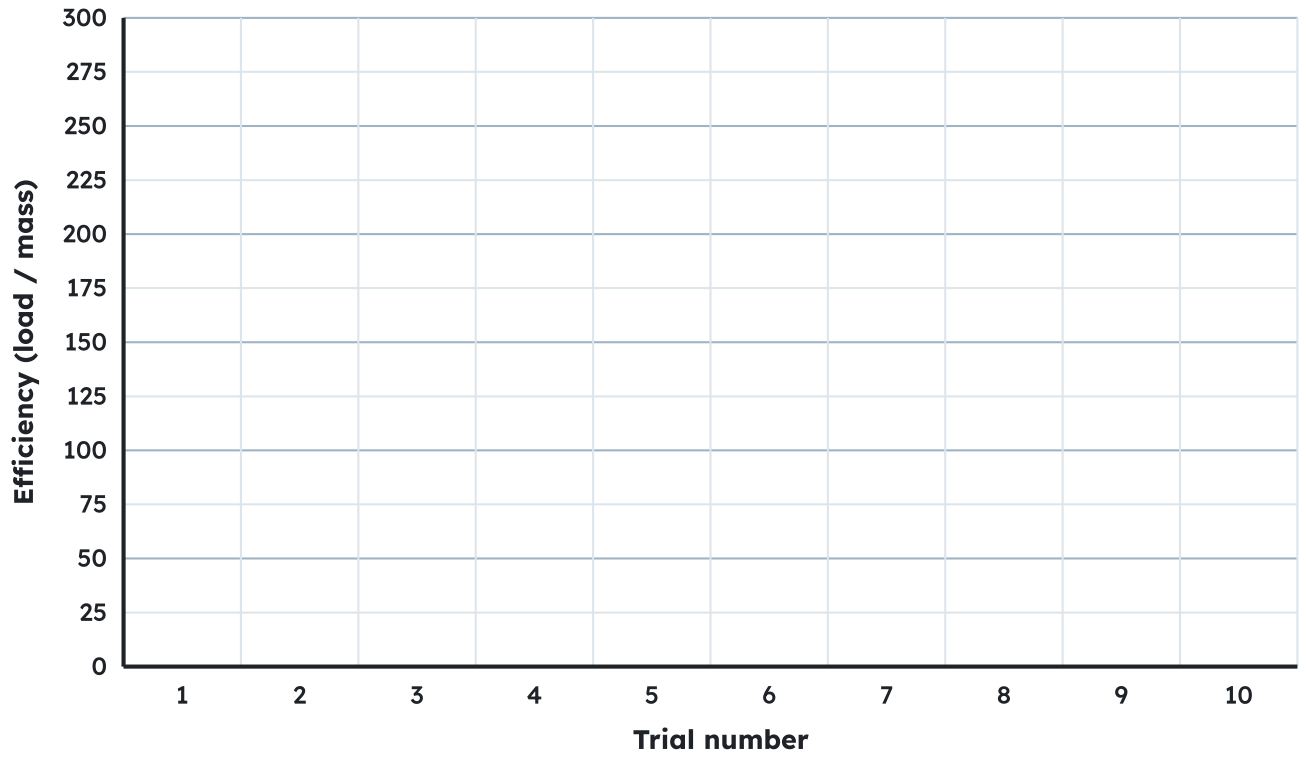
Next time I will change:

_____

_____

My Best-Score Graph

After every trial, put a dot at your efficiency. Connect the dots. Is your line going up?



My best score

Efficiency: _____ Trial #: _____

Mass: _____ g Load: _____ g

My goal line

Draw a dashed line across the graph at your goal.

Goal: _____

The Judge practice bridges in Lab I2 scored from 120 to 200.

What change made my score go up the most?

☐ Version 2 Builder badge: Version 2 tested and graphed.

What Went Wrong?

Find your problem. Read why it happens. Try the fix, then check the box. Engineers call this **troubleshooting**.

Problem	Why it happens	Try this	Tried
It twisted or leaned	The two trusses do not match, or there are not enough cross braces.	Use the jig blocks. Add braces at both ends. Look from the end after each brace.	☐
A slanted side snapped	Compression bends long thin pieces.	Bundle more strands there, or add diagonals so each piece is shorter.	☐
The bottom chord snapped	Tension pulled apart a weak spot or a short splice.	Use full-length strands. Overlap splices a lot and spread them out.	☐
A joint pulled apart	Too little glue, or the glue was not dry.	One small dab where pieces touch. Hold 30 seconds. Dry a full day.	☐
Platform under 20.0 cm	Measured from the table under the gap.	Measure from where the ends sit on the supports. Aim for 20.5 to 21 cm.	☐
Chain hit the bridge	Braces blocked the middle.	Leave a 3 cm gap. A pencil should drop straight through.	☐
Wider than 7.5 cm	Trusses spread apart while glue dried.	Jig blocks about 5.5 cm wide. Measure before the glue dries.	☐
Too heavy	Glue blobs, or extra strands where nothing broke.	Wipe blobs. Take strands out of parts that had no damage.	☐

My own problem

What happened:

Why I think it happened:

My fix, and did it work?

AT-HOME MINI BUILD CHALLENGE 1 · AFTER BUILD SESSION 1

Triangle Tower

Earns the **Triangle Tamer** badge. You build; a grown-up watches.

Goal: Prove that triangles are stronger than squares, then build the tallest tower that holds a book.

You need

- [ ] 30 toothpicks (or 5 cm pasta pieces)
- [ ] Mini marshmallows, gumdrops, or play dough balls
- [ ] A paperback book
- [ ] Ruler
- [ ] A timer

Steps

1. Build a square from 4 toothpicks. Press gently on one corner. What happens?
2. Add one diagonal toothpick. Press again.
3. Now build a cube (12 toothpicks). Push the top sideways. Add diagonals to its sides until it stops wobbling.
4. Set the timer for 10 minutes. Build the tallest tower you can that holds the paperback book on top for 10 seconds.
5. Measure the height and count the toothpicks. Fill in the table.

What I built	Toothpicks used	Height (cm)	Held the book? What happened?
Square			
Square + 1 diagonal			
My tower			

Where did you add triangles to your tower? Why did they help?

Find 2 triangles in real life (a bridge, a swing set, a roof). Where were they?

Draw your build, or tape a photo here

AT-HOME MINI BUILD CHALLENGE 2 · AFTER BUILD SESSION 2

Straw Bridge

Earns the **Straw Bridge Engineer** badge. You build; a grown-up watches.

Goal: Build a bridge from straws and tape that crosses a 20 cm gap and holds the most pennies for the fewest straws.

You need

- [ ] 10 plastic or paper drinking straws
- [ ] Tape
- [ ] 2 stacks of books the same height
- [ ] A small paper cup
- [ ] Up to 100 pennies
- [ ] Ruler, scissors

Steps

1. Set the book stacks 20 cm apart.
2. Draw your bridge first in the box below. Use triangles!
3. Build a bridge with 10 straws or fewer. It must rest on top of the books.
4. Set the cup on the middle of the bridge. Add pennies one at a time until it sags to the table or breaks.
5. Score it: pennies held ÷ straws used. Then change ONE thing and try again.

My straw bridge plan

	Straws used	Pennies held	Pennies per straw
Try 1			
Try 2			

What ONE thing did you change for Try 2? Did your score go up?

How is this like the real Bridge-A-Roni score?

Try 2: draw your better bridge, or tape a photo here

AT-HOME MINI BUILD CHALLENGE 3 · AFTER BUILD SESSION 3

Column Crusher

Earns the **Column Crusher** badge. You build; a grown-up watches.

Goal: Find out which column shape holds the most push (compression). Every column uses one card, so they all have the same mass.

You need

- [ ] 3 index cards (or 3 strips of cereal-box cardboard the same size)
- [ ] Tape
- [ ] A stiff piece of cardboard or a light tray
- [ ] Paperback books or magazines, all about the same
- [ ] Ruler

Steps

1. Fold card 1 into a **triangle** tube, card 2 into a **square** tube, and roll card 3 into a **circle** tube. Tape each seam.
2. Stand one column up on the floor. Lay the cardboard on top.
3. Stack books on the cardboard one at a time, gently, until the column crushes.
4. Count the books it held. Test the other two shapes the same way.
5. Predict first, then test!

Safety: test on the floor, keep the stack low, and keep toes out from under the books.

Column shape	My prediction (books)	Books it held
Triangle		
Square		
Circle		

Which shape held the most? Why do you think so?

Which parts of your pasta bridge get squeezed like these columns?

Draw your build, or tape a photo here

Tournament-Day Checklist

Partners bring one pasta-and-glue bridge they built before the tournament plus safety glasses, then have 10 minutes to have it massed and measured, set it on the tester, and pour sand into a hanging bucket for up to 3 minutes.

Pack the night before

- ☐ Our ONE bridge, in a flat padded box
- ☐ Safety glasses for both partners
- ☐ This logbook (your Version 2 numbers are inside)
- ☐ A pencil
- ☐ A good night's sleep!

Know your job

Pourer
Pours the sand for up to 3 minutes.

_____

Steadier
Keeps the bucket from swinging. Never holds it up.

_____

Score sheet check (all YES = Tier 1)

- ☐ A single structure made of ONLY pasta and glue
- ☐ Sits on top of the test supports (45.0 cm span)
- ☐ Loading block rests freely at the center
- ☐ Loading block at least 20.0 cm above the support tops
- ☐ Opening lets the bolt and chain pass through
- ☐ Before loading, bridge is on or above the support tops
- ☐ Width 0 cm to 7.5 cm
- ☐ We did it with no outside help

At the tester

- ☐ Glasses on, both of us
- ☐ Hand over the bridge to be massed and measured
- ☐ Ends rest only on the support tops
- ☐ Block in the center, chain through the gap
- ☐ Bucket hangs free, not touching the floor
- ☐ Pour steadily; stop at the break

My tournament result

Mass: _____ g Load: _____ g

Efficiency: _____ Tier: ___

What I will tell next year's team

_____

_____

_____

☐ Rule Checker badge: every score sheet line YES before the tournament.

Our tournament bridge: draw it or tape a photo

Build Missions

One mission for each build session, plus one before the tournament. When you finish, show a coach or your Event Captain for initials, then color in the badge.

When	Mission	Badge	Done	Initials
Build session 1 At home after	Triangle Tower challenge (p. 13).	Triangle Tamer	☐	
Build session 1 At home after	Finish your side truss and log it in the Build Log (p. 8). Label T and C (p. 4).	Truss Builder / Force Finder	☐	
Build session 2 At home after	Straw Bridge challenge (p. 14).	Straw Bridge Engineer	☐	
Build session 2 Lab I2	Mass budget (p. 5), then compare it to your real mass.	Mass Budgeter	☐	
Build session 3 At the session	Log your load test on a Trial Log (p. 9).	Load Tester	☐	
Build session 3 At home after	Column Crusher challenge (p. 15).	Column Crusher	☐	
Before tournament Version 2	Draw Version 2 (p. 7), build it, test it, and graph it (p. 11). Score sheet all YES (p. 16).	Version 2 Builder / Rule Checker	☐	

Tournament day: Sat Mar 13

- ☐ Bridge in a flat padded box
- ☐ Safety glasses for both partners
- ☐ This logbook

- Run the score sheet check the night before.
- Decide who is Pourer and who is Steadier.
- Stop pouring the second it breaks.

Engineer's notes: ideas, questions, and things to bring

My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Bridge Engineer**! Your Team Badge Passport has the same badges.



Triangle Tamer

Triangle Tower challenge

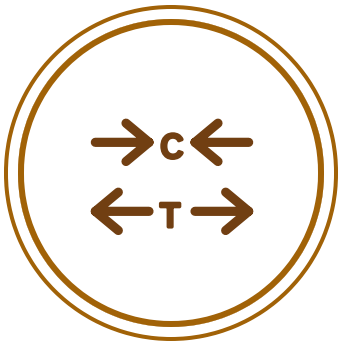
Initials: _____



Truss Builder

Finish a side truss at home

Initials: _____



Force Finder

Label T and C on the truss

Initials: _____



Straw Bridge Engineer

Straw Bridge challenge

Initials: _____



Mass Budgeter

Estimate your bridge's mass

Initials: _____



Column Crusher

Index Card Column challenge

Initials: _____



Load Tester

Log a real load test

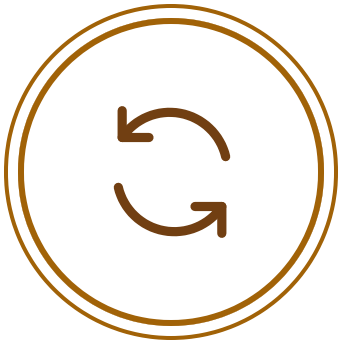
Initials: _____



Rule Checker

All YES on the score sheet check

Initials: _____



Version 2 Builder

Redesign and graph your scores

Initials: _____